Laeeq Aslam

Machine Learning Engineer | AI & Edge Computing | Sustainable Systems

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Professional Summary

Machine learning researcher with a PhD in Control Science and Machine Learning, specializing in **physics-informed models** and **AI for sustainable systems**. Experienced in **time-series forecasting**, **edge computing**, and **optimization algorithms**. Eager to advance cutting-edge research in AI and renewable energy systems, with a strong record of first-author publications and interdisciplinary collaborations. Seeking to contribute to impactful AI-driven solutions in **energy systems** and **climate-related challenges**.

Publications & Research

Published **9 journal** articles, **4 arXiv preprints**, **3 conference papers,** and **4 under review,** with 31 total citations. Recent publications as **first author** are as follows,

- ***Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction (Computers & Electrical Engineering, 2025).*** *Developed a novel spatio-temporal model for wind speed prediction, achieving more than 9.5% improvement in accuracy compared to Mixformer, showcasing the integration of physics-based knowledge in machine learning.*
- ***A Shallow Hybrid Model with Dynamic Bayesian Optimisation for Wind Speed Prediction on Memory-Constrained Devices. (Computers & Electrical Engineering).*** *This work proposes the D-TPE-SHM model for wind speed prediction on memory-constrained devices, combining Fast-ICA, Adaptive LASSO, and Concurrent Recurrent Temporal Module to capture temporal patterns, with hyperparameters optimized using the dynamic D-TPE algorithm, which adjusts thresholds based on gradient feedback to improve convergence speed and model performance. The model outperforms existing methods in accuracy and efficiency, making it suitable for real-time deployment on devices such as Jetson Nano.*
- **Dynamic Optimization of Recurrent Networks for Wind Prediction on Edge Devices.** ***(IEEE Access, 2025).*** *Developed adaptive Simulated Annealing with Memory-based Rejection (aSAR) for hyperparameter optimization, co-tuning LSTM, GRU, and TCN with dynamic cooling and a rejection memory that avoids revisiting weak solutions. Using nine objective functions and terrain-specific tuning, the method outperformed iTransformer, Flowformer, IDBO-VTGA, PI-LSTM, and 1D CNN-LSTM, achieving up to 98.75% smaller models and up to 54.17% higher predictive accuracy.*
- **Hardware-Centric Exploration of the Discrete Design Space in Transformer–LSTM Models for Wind Speed Prediction on Memory-Constrained Devices.** ***(Energies,2025).*** *This study integrates hardware considerations into the optimization process for a hybrid transformer-LSTM model, with a focus on achieving real-time performance. Models are trained on a server, pushed to hardware via gRPC over a local network, and tested on the Jetson Nano device. This approach ensures that hyperparameters are optimized while considering the real-time performance of the model on edge devices, such as the Jetson Nano.*
- **Integrating Physics-Informed Vectors for Improved Wind Speed Forecasting with Neural Networks.** ***(Asian Control Conference, 2024).*** *This research improves wind speed forecasting by integrating Physics-Informed Vectors (PIV) into LSTM and TCN models. The approach uses atmospheric*

data from multiple directions to calculate PIV, which improves model accuracy. A custom hybrid loss function accelerates model convergence. The method shows significant improvements in prediction performance across datasets from multiple countries.

Under Review

- **A Distribution Matching Framework for Zero-Shot Wind Speed Prediction on Edge Devices in Unseen Locations.** ***(Under Review, Engineering Applications of Artificial Intelligence).*** *This work proposes a Distribution Matching Framework (DMF) for zero-shot wind speed prediction on resource-constrained edge devices, utilizing pre-trained location-specific recurrent neural networks with attention (RNNA) across 460 locations. By employing Empirical Distribution Matching (EDM) and Multi-Objective Loss Optimization (MOLO), the framework ensures efficient model deployment for small-scale wind turbines, achieving robust performance with a compact model size of 7.3 MB. The method demonstrates superior performance with low Jensen-Shannon divergence (below 0.0015) across five unseen locations.*
- ***Physics-Informed Ensemble Learning for City-Scale Temperature Prediction During Thermal Extremes PIEL-NET (Under Review, Urban Climate)****. This work fuses a Physics-Terrain agent, a spatio-temporal ConvLSTM agent trained with mean absolute error, and a regime-specialized agent trained with Regime Adaptive Focal Loss. Applies hierarchical adaptive fusion over nine reanalysis cells to predict 2 m air temperature at the urban core. Cuts error by 69.2% below minus 7.4 degrees Celsius in Calgary's extreme cold and reaches a Dice score of 0.89 for warnings.*
- ***PiDR-Net: A Perception-Informed Deep Recurrent Network for Short-Term Wind Speed Predictions (Submitted, Applied Soft Computing)****. This work combines an ensemble deep random vector functional link with skip connections for spatial perception and LSTM for temporal modeling. Uses a dynamic perception-informed loss with a fuzzy variance module to stabilize training. Reduces mean squared error by up to 51.1% versus the strongest competing state-of-the-art models and improves long-horizon accuracy during volatile periods.*
- ***Physics-Informed Extreme Wind Speed Prediction with a Multi-Gated Convolutional Recurrent Network VORTEX-Net (submitted to Applied Energy).*** *This work introduces a physics-based volatility proxy from pressure-gradient force discrepancy to guide predictions. Uses a Membership-Gated Convolutional Recurrent Module and a Differentiable Evolutionary Mixture Layer. A volatility-adaptive loss raises accuracy during high-impact events. On NASA POWER sites with 24-hour inputs for 6-hour ahead prediction, lowers RMSE by 8.40% and MAE by 9.30% on Jiuquan, and cuts MSE by 18.1% in the highest volatility quartile. Improves wind power density forecasting for grid operations.*

Work Experience

Bond and Built Pvt Ltd, Pakistan

AI/ML Consultant - Edge Computing & Computer Vision | July 2024 – March 2025

- Led the development of a real-time **footwear analytics system** deployed across urban Pakistan.
- Scaled system to process **50,000+ daily inferences** with **60+ concurrent streams**.
- Delivered actionable **market intelligence** for strategic material investments.

Central South University, China

PhD Researcher – Machine Learning & AI for Sustainable Systems | Sept 2020 – Present

- Developed **Physics-Informed Spatio-Temporal Network (PISTNet)**, achieving **more than 9.5% improvement in SMAPE** against state-of-the-art models, including iTransformers, PatchTST, and Mixformer for wind speed forecasting.

- Proposed **Adaptive Simulated Annealing with Memory-based Rejection for Recurrent Neural Networks Discrete Hyper-Parameter Space optimization**, reducing **MAE by 14.5%** against baseline models (LSTM, TCN) in time-series forecasting.
- Contributed to the Key **R&D Program of Hunan Province, Project 2020WK2007**, focusing on optimizing energy integration and improving grid performance for wind energy storage.

DLISION, Pakistan

Machine Learning Engineer – Computer Vision & AI Deployment | May 2021 – Sept 2022

- Led **computer vision projects** focused on **semantic segmentation and real-time object detection**.
- Optimized **CNN-UNet & Swin UNet models**, improving **segmentation accuracy by 3% against their baseline architectures** using **Focal Loss**.

International Islamic University, Pakistan

Higher Education Research Assistant – AI for Healthcare | March 2019 – Feb 2020

- Applied **Generative Adversarial Networks (GANs)** for **breast cancer detection**, improving classification accuracy.
- Conducted data augmentation techniques that **increased dataset diversity** and **enhanced model robustness**.

iUSE School of Engineering and Management Sciences, Pakistan

Lecturer & Lab Engineer – Embedded Systems & AI | Sept 2013 – Feb 2019

- Taking undergraduate courses related to **Embedded Systems**, **Programming,** and **Discrete Signal Processing**.
- Designed and developed an **Instrumentation & Measurement Lab** for **sensor-based data acquisition in MATLAB**.
- Led **student research projects** on **energy management, accident alerts, and secure electronic voting**.

Education

- PhD in Control Science & Machine Learning | Central South University, China | 2020 – Present.
- MS in Electronic Engineering| International Islamic University, Pakistan | 2015 – 2017

Technical Skills

- **Machine Learning & AI:** Deep Learning | Computer Vision | Time-Series Analysis | Predictive Modeling.
- **Programming & Tools:** Python | MATLAB | C++ | TensorFlow | PyTorch | Keras | OpenCV | Edge Impulse.
- **DevOps & Deployment:** Docker | AWS (EC2)| Git | Triton Inference Server.
- **Hardware & IoT:** Raspberry Pi | NVIDIA Jetson | Arduino Nano BLE Sense.
- **Data Analysis & Optimization:** Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering

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Key Projects & Research Contributions

- Deployed a **real-time edge-to-cloud footwear analytics pipeline** (YOLOv5 → DenseNet-121), processing **50k+ daily inferences** across urban Pakistan with **60+ concurrent streams.**
- **AI-Based Sports Analytics:** Enhanced **real-time player segmentation** using **U-NET models**, improving **inference time by 15%**.
- Developed and published the **keras-swin-unet** PIP package for satellite imagery segmentation and the **TimeMesh** library for efficient time series data preprocessing.

Awards & Honors

- Gold Medalist – MS in Electronic Engineering | International Islamic University.
- Hunan Provincial Government Funding for Research & Development (**Project 2020WK2007)**.
- Chinese Government CSC Scholarship for PhD Studies.

Professional Memberships & Certifications

- Registered Engineer, Pakistan Engineering Council (ELECTRO/22837)
- Member, Institute of Electrical and Electronics Engineers (IEEE)